Ubuntu Server and Windows Server Evaluation

INSTALLATION GUIDE

Prepared by: Arianna C. Surbano

Program: Bachelor of Science in Information Technology

Instructor: John Randolf M. Penaredondo, MIT

Date: August 15, 2026

1. Project Overview

This project is about installing and configuring Ubuntu Server and Windows Server using virtual machines. It also includes BIOS vs UEFI comparison, Ubuntu boot process, and comparison of server operating systems.

2. Objectives

• Install Ubuntu Server.
• Configure and verify the server.
• Install Windows Server Evaluation.
• Compare BIOS and UEFI.
• Create an Ubuntu boot process flowchart.
• Compare Windows Server, Ubuntu Server, and Rocky Linux.

3. Software Requirements

Software	Purpose
VirtualBox	Create virtual machines
Ubuntu Server ISO	Ubuntu installation
Windows Server ISO	Windows Server installation
Draw.io	Flowchart
Microsoft Word	Documentation

4. Virtual Machine Specifications

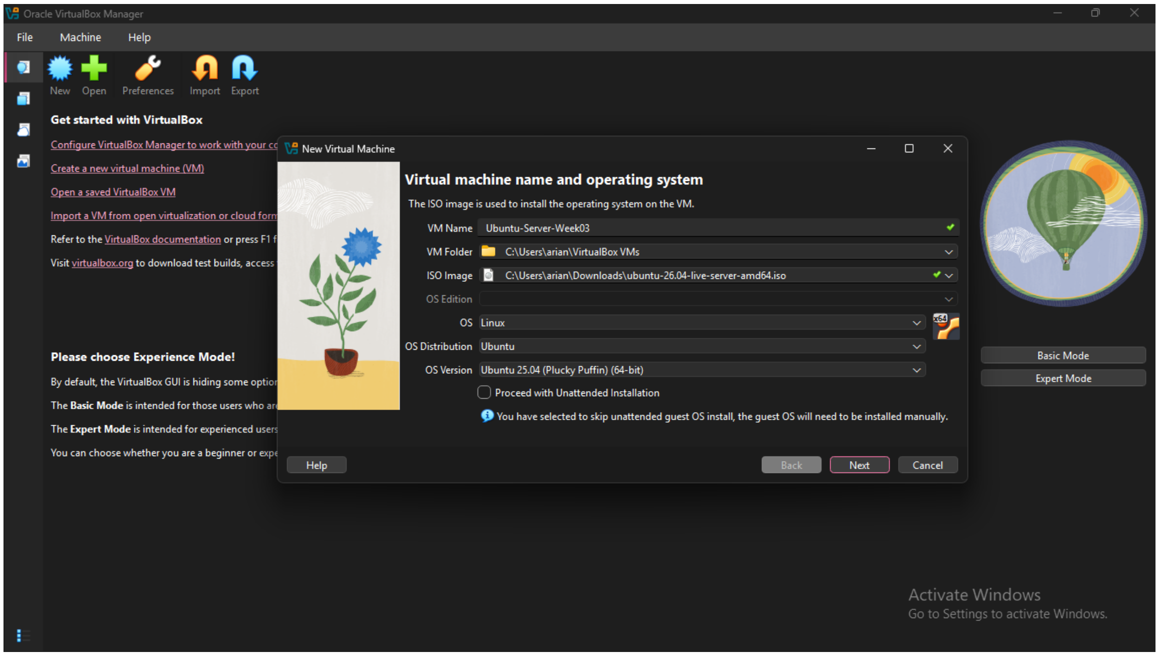
Ubuntu Server
• OS: Ubuntu Server
• RAM: 4 GB
• Network: NAT
• Hostname: ariannasurbano
• IP Address: 10.0.2.15

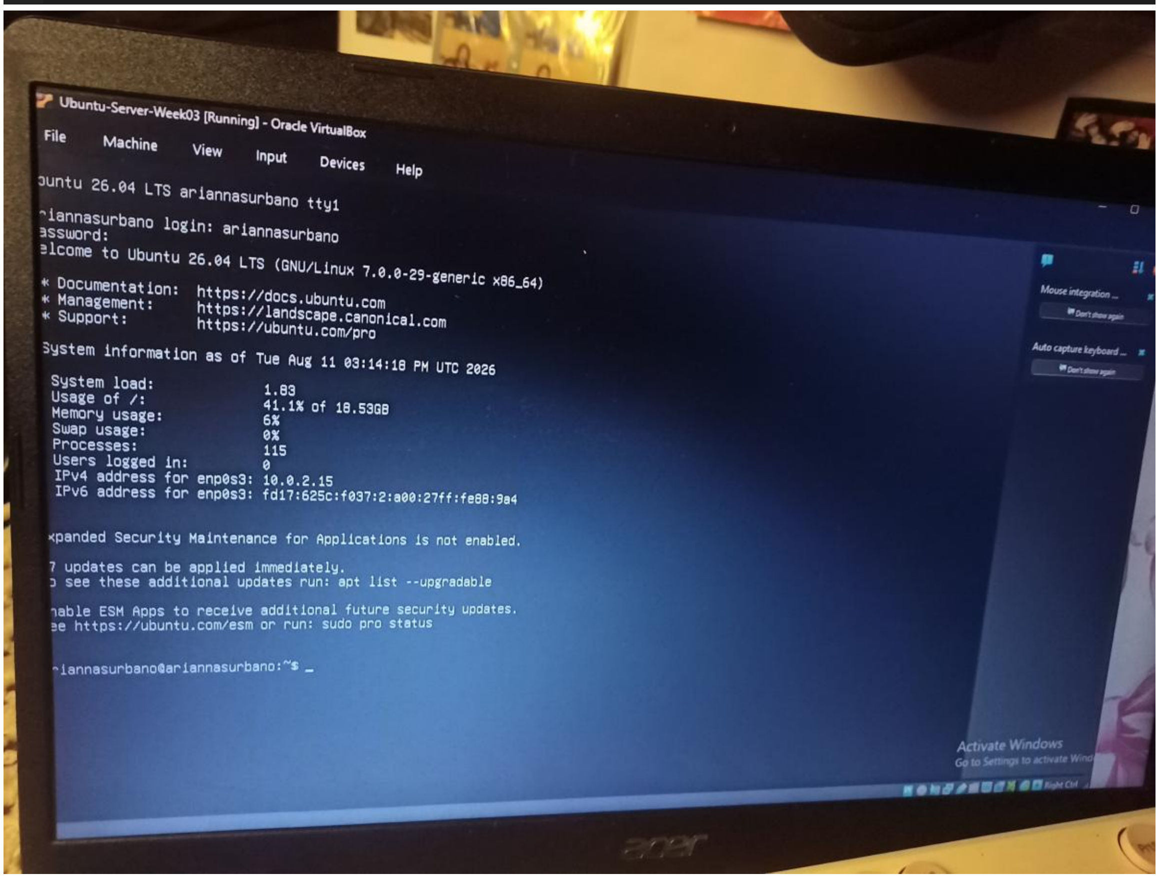
Windows Server
• OS: Windows Server Evaluation
• RAM: 4 GB
• CPU: 2 CPUs
• Storage: 50 GB
• Network: NAT

5. Step-by-Step Installation

Ubuntu Server

1. Create a new VM in VirtualBox.
2. Select Ubuntu Server ISO.
3. Set RAM, CPU, and storage.
4. Install Ubuntu Server.
5. Create a user and password.
6. Login to the server.

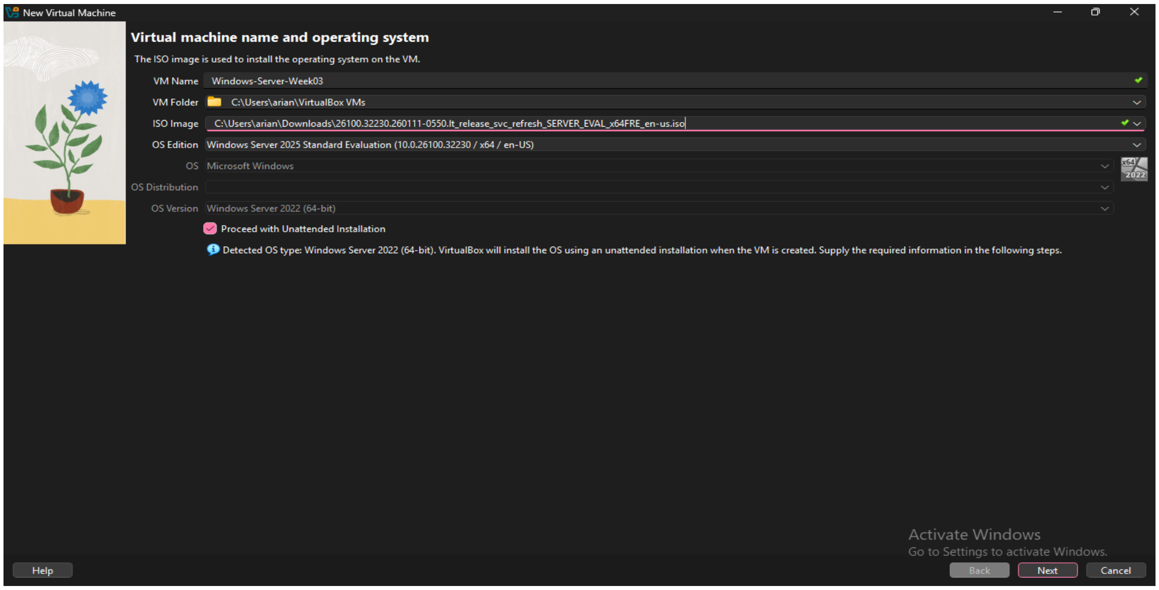




Windows Server

1. Create a new VM in VirtualBox.
2. Select Windows Server Evaluation ISO.
3. Set RAM to 4 GB.

4. Set storage to 50 GB.
5. Start the VM.
6. Install Windows Server.
7. Create Administrator password.
8. Login.
9. Set the computer name.

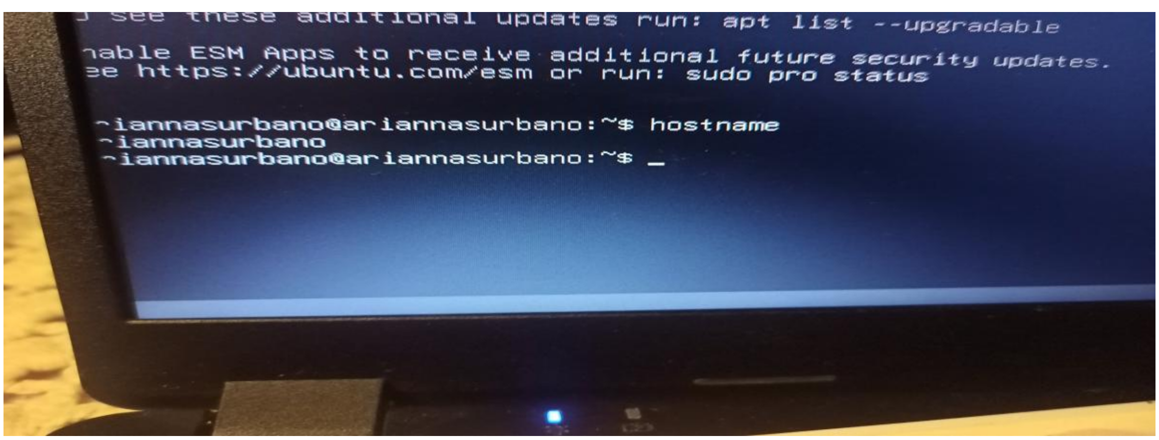


6. Configuration and Verification

Hostname

hostname

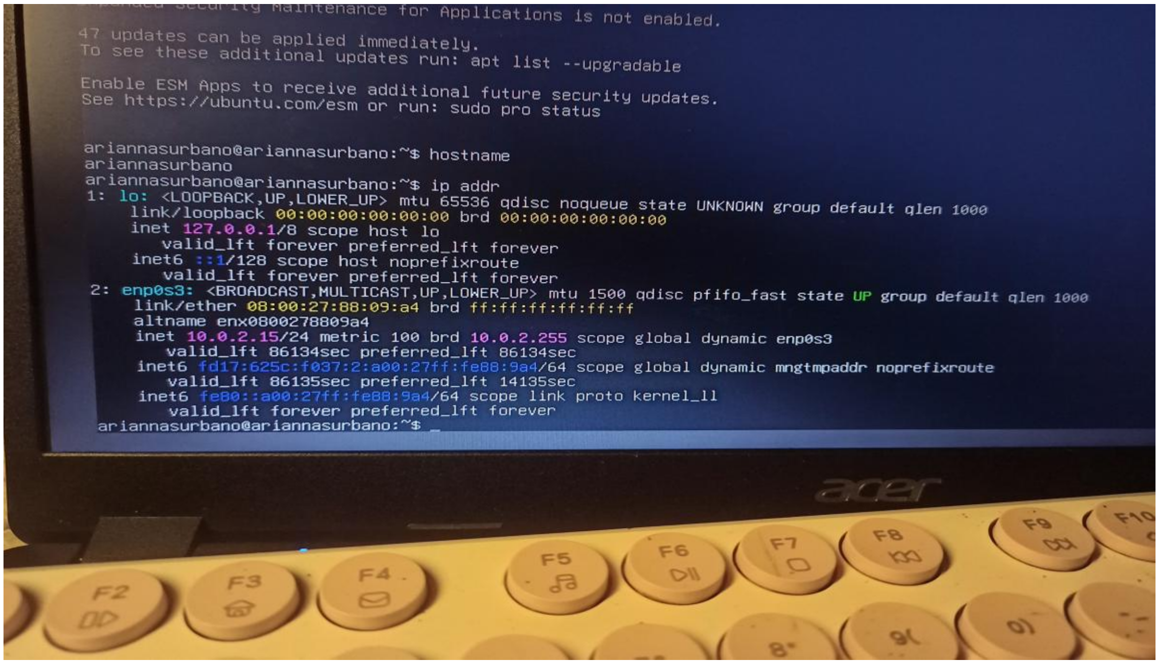
Result: ariannasurbano



IP Address

ip addr

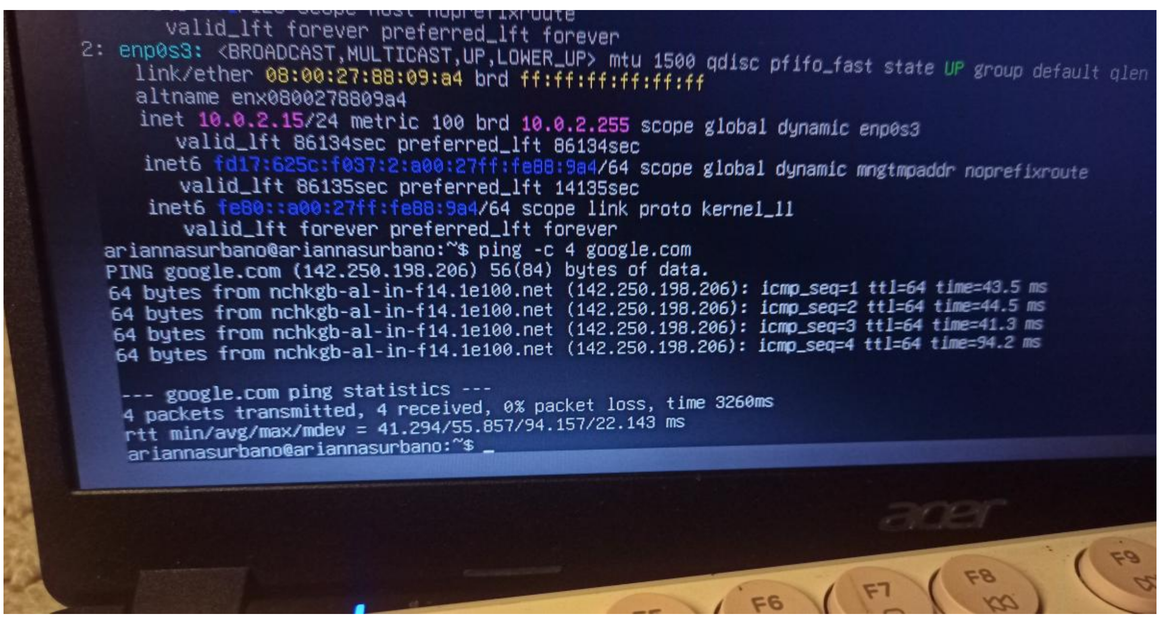
Result: 10.0.2.15



Internet

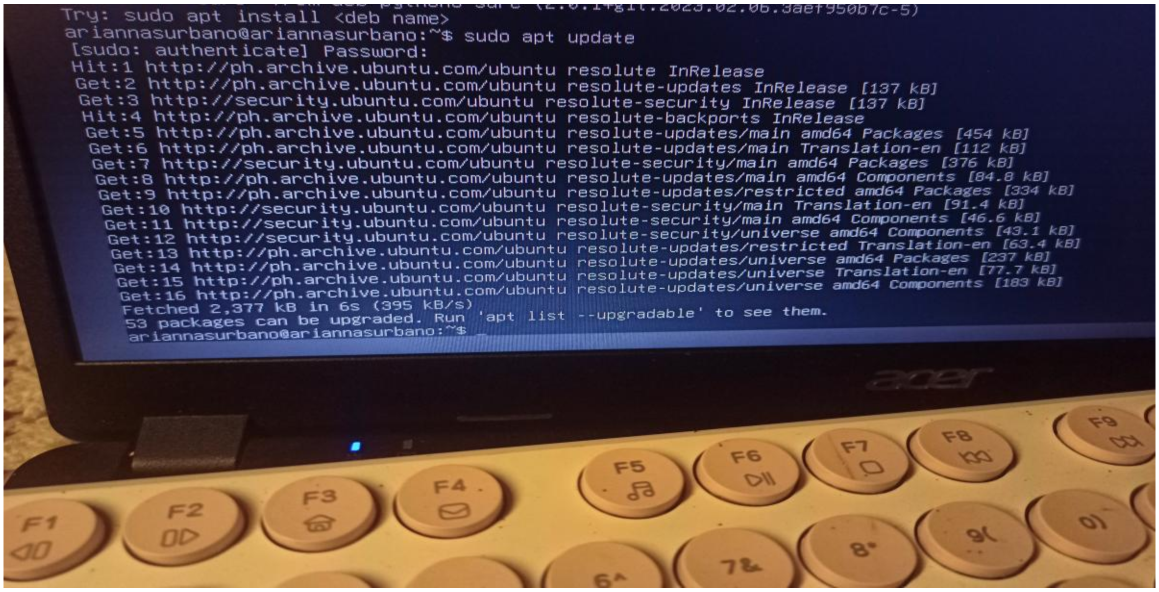
ping -c 4 google.com

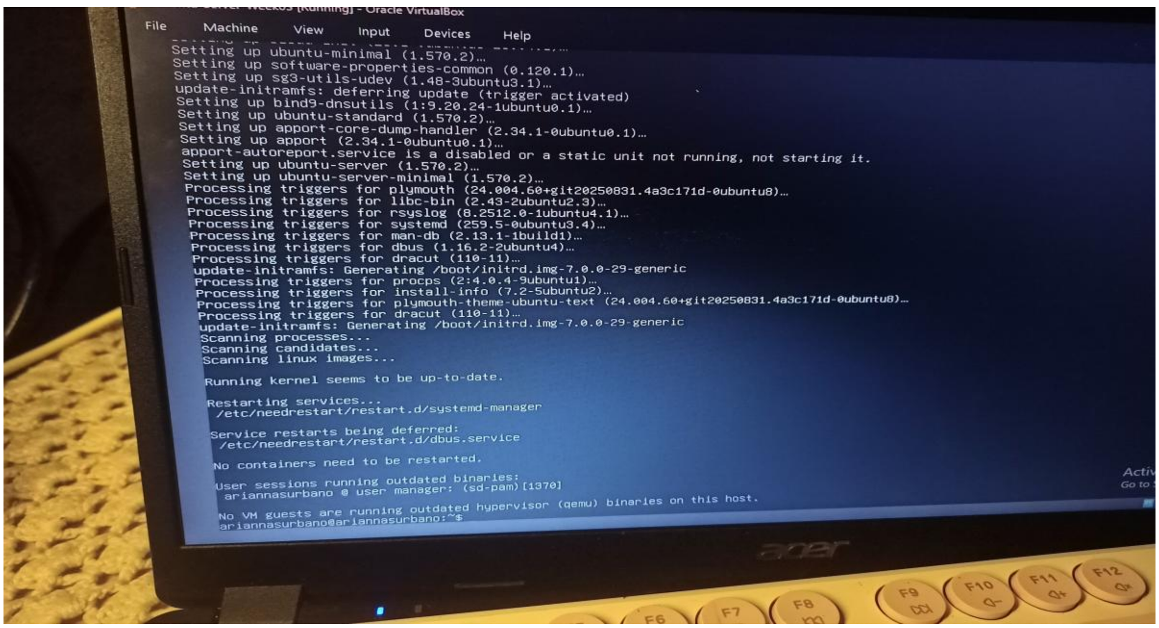
Result: Successful connection



apt upgrade -y

Result: Update completed

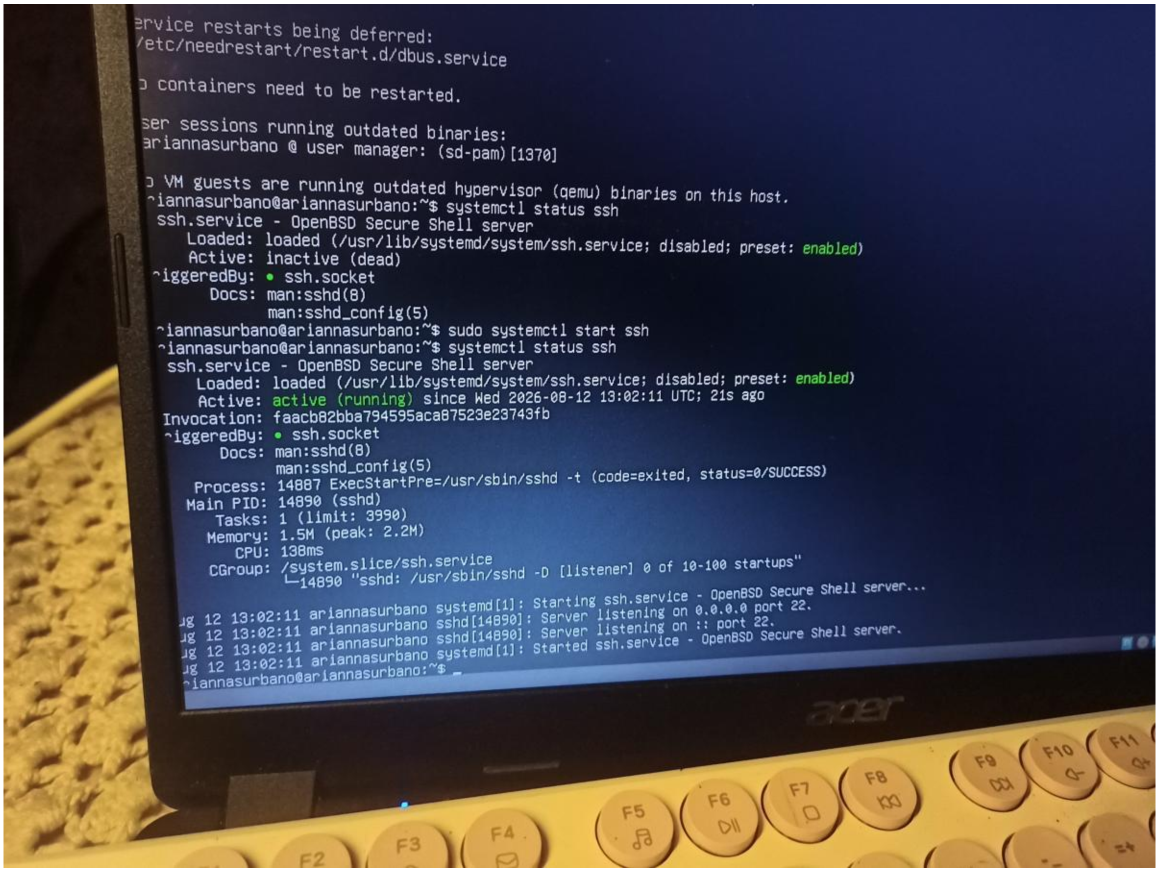




SSH

systemctl status ssh

Result: active (running)

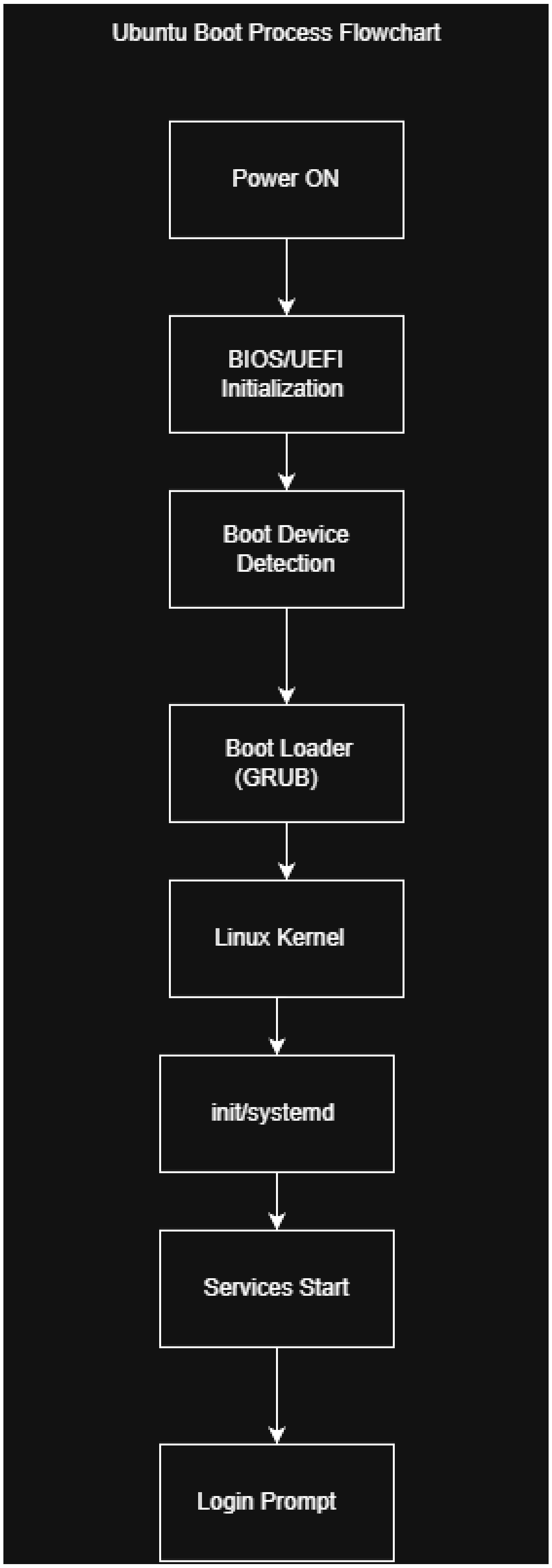


7. BIOS vs UEFI

Category	BIOS	UEFI
Definition	BIOS (Basic Input/Output System) is older firmware that initializes hardware and starts the operating system.	UEFI (Unified Extensible Firmware Interface) is a modern firmware interface designed to replace traditional BIOS.
Boot Process	Uses a traditional boot process where BIOS initializes hardware and loads the boot code from the disk's MBR.	Initializes hardware and loads a boot manager from the EFI System Partition (ESP).
Maximum Disk Support	Traditionally limited by the MBR partitioning scheme to about 2.2 TB for a boot disk.	Works with GPT, allowing support for disks much larger than 2 TB. GPT's theoretical limit is far beyond today's typical disk sizes.
Partition Style	Commonly uses MBR (Master Boot Record).	Commonly uses GPT (GUID Partition Table). A

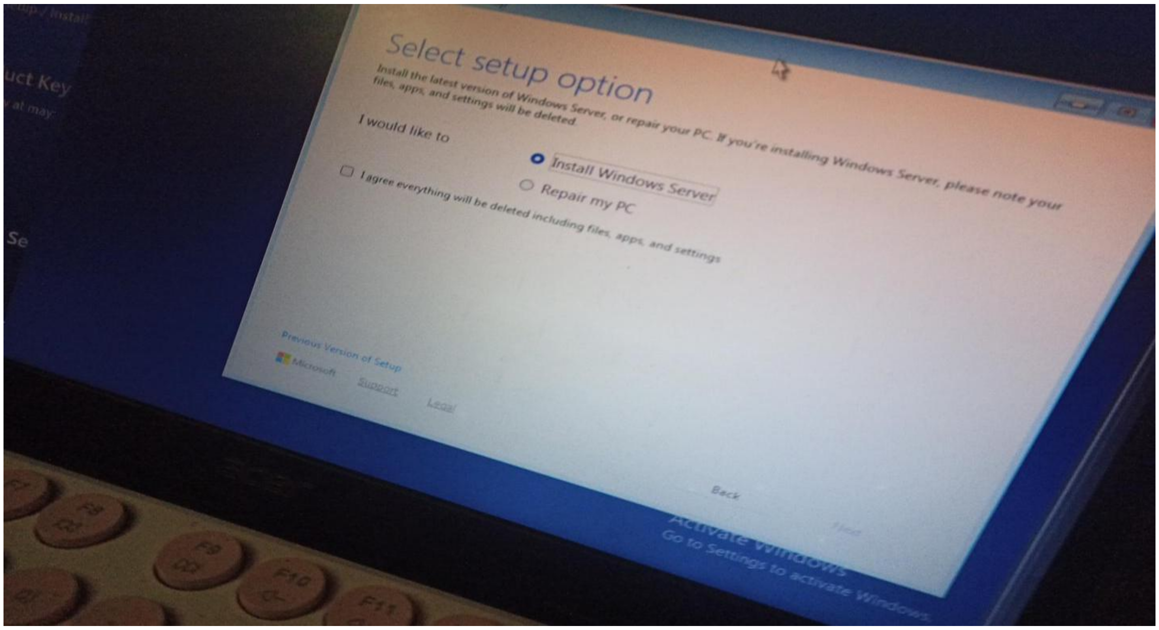
		UEFI Windows boot disk must use GPT.
Security Features	Has limited built-in boot security and does not provide UEFI's Secure Boot mechanism.	Supports Secure Boot, which verifies trusted boot software before allowing the system to start.
Speed	Generally has a slower and more limited boot process.	Generally provides faster boot and resume times.
Advantages	Simple, widely compatible with older hardware and operating systems, and useful for legacy systems.	Faster startup, supports large disks and more partitions, provides modern security features, and offers greater flexibility.
Disadvantages	Limited by older technology, MBR restrictions, and fewer modern security features.	Can be more complex to configure and may have compatibility issues with very old operating systems or hardware.
Modern Usage	Mostly used for legacy computers and compatibility with older systems.	Used on most modern computers and is the preferred firmware standard for current systems.

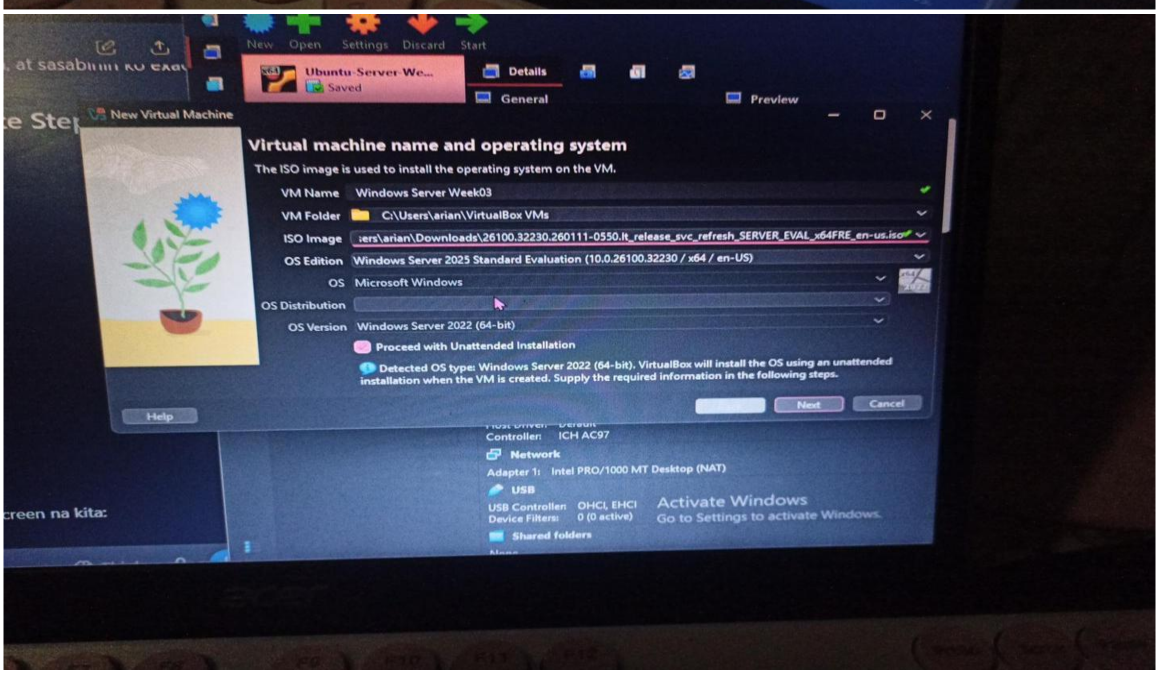
8. Ubuntu Boot Process

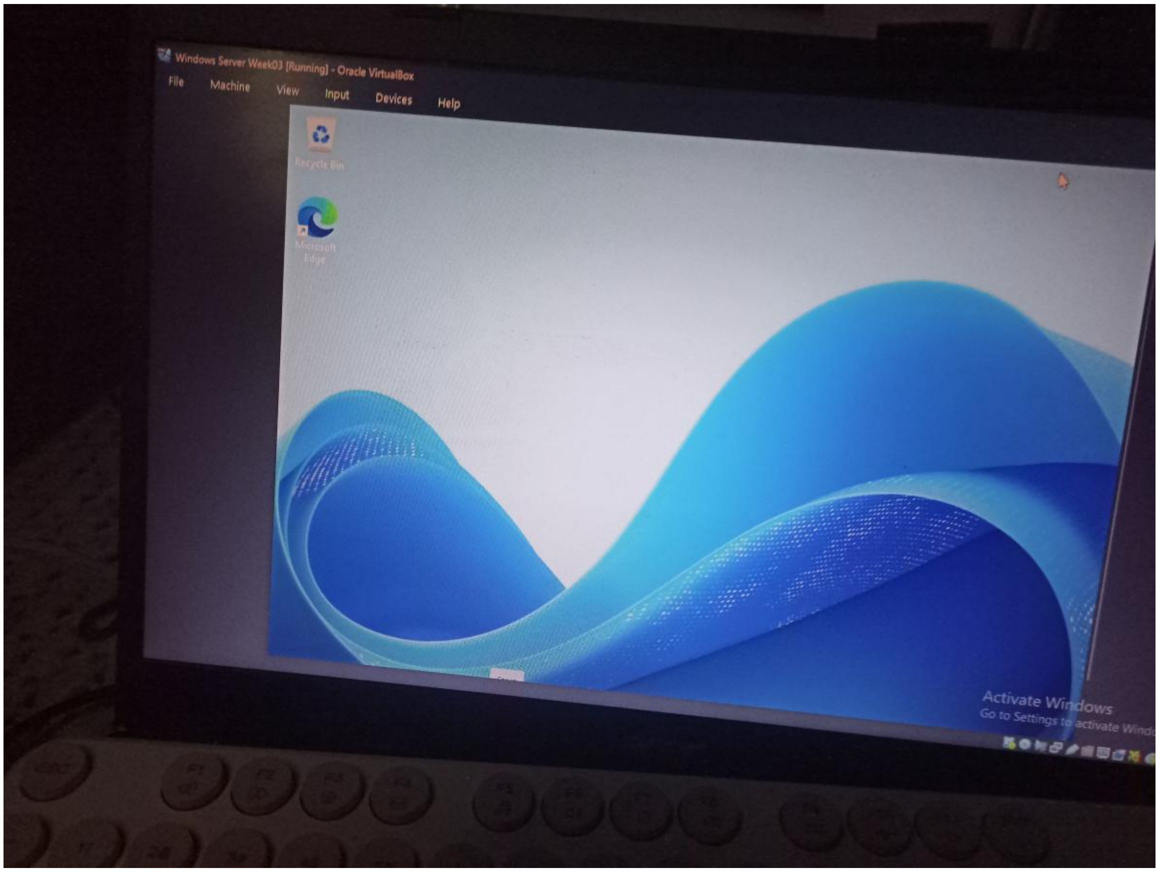


9. Windows Server Installation Summary

Windows Server Evaluation was installed using a virtual machine. The VM was configured with 4 GB RAM, 2 CPUs, and 50 GB storage. The Windows Server Evaluation ISO was used for installation. An Administrator password was created, the server was successfully logged in, and a computer name was assigned.







10. OS Comparison

Category	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial	Free/Open Source	Free/Open Source
User Interface	GUI available	Mostly CLI	Mostly CLI
Package Management	Windows tools	APT	DNF
Security	Strong	Strong	Strong
Performance	Good	Good	Good
Enterprise Use	Windows networks, Active Directory	Web/cloud servers	Enterprise Linux servers
Advantages	Easy for Windows users	Free and flexible	Stable and secure
Disadvantages	Can require licensing	Requires Linux knowledge	Requires Linux knowledge

11. Problems Encountered

Problem 1: Ubuntu update had a repository error.

Problem 2: Windows Server VM initially failed to boot because the ISO was not properly mounted.

12. Solutions

Ubuntu: The package repository was corrected and the update was run again.

Windows Server: The correct Windows Server ISO was selected and mounted in VirtualBox Storage.

13. Lessons Learned

I learned how to install and configure server operating systems using virtual machines. I also learned how to check IP addresses, test internet connection, update Ubuntu, verify SSH, install Windows Server, and troubleshoot basic installation problems.

14. References

- Microsoft. (2023). UEFI Firmware Requirements. Microsoft Learn. https://learn.microsoft.com/en-us/windows-hardware/design/device-experiences/oem-uefi
- Microsoft. (2021). Windows Support for Hard Disks Exceeding 2 TB. Microsoft Learn. https://learn.microsoft.com/en-us/troubleshoot/windows-server/backup-and-storage/support-for-hard-disks-exceeding-2-tb
- Microsoft. (2026). Secure Boot. Microsoft Learn. https://learn.microsoft.com/en-us/windows-hardware/design/device-experiences/oem-secure-boot
- UEFI Forum. (2024). UEFI Specifications. Unified Extensible Firmware Interface Forum. https://uefi.org/specifications
- Canonical. (2026). Get Ubuntu Server. Canonical Ltd. https://ubuntu.com/download/server
- Microsoft. (2026). Windows Server Pricing & Licensing Guide. Microsoft Corporation. https://www.microsoft.com/en-us/windows-server/pricing
- Canonical. (2026). Ubuntu Server Documentation. Canonical Ltd. https://canonical-ubuntu-server.readthedocs-hosted.com/en/latest/
- Rocky Enterprise Software Foundation. (2026). Rocky Linux Documentation & RHEL Compatibility. Rocky Enterprise Software Foundation. https://docs.rockylinux.org/